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Foreign Key (FK)

A Foreign Key (FK) is an attribute (or a set of attributes) in one table that refers to the Primary Key (or another unique key) of another table. foreign keys connect tables ,enforce relationships , prevents invalid data (refernetial integrity , fk must be equal to a valid pk value or allowed null values , Maintaining Consistency

Foreign Keys ensure that related tables remain synchronized.

Foreign Key Constraints & Actions

CASCADE

ON DELETE CASCADE
When a parent row is deleted,
all matching child rows are automatically deleted.

ON UPDATE CASCADE
Suppose
StudentID changes
101 → 201
The database automatically updates all Foreign Keys.

RESTRICT (or NO ACTION)

This is the safest option.
If child rows exist,
the parent row cannot be deleted (or updated, depending on the constraint).

SET NULL

When the parent row is deleted,
the Foreign Key value becomes NULL.

SET DEFAULT

When the parent row is deleted,
the Foreign Key is set to a predefined default value.

Example 1 — Student and Course

Student

StudentID	Name
101	Arpit
102	Aadya

Enrollment

EnrollmentID	StudentID	Course
1	101	DBMS
2	102	OS

Foreign Key
Enrollment.StudentID
↓
references
↓
Student.StudentID

Creating Parent Table

```
CREATE TABLE Student
(
    StudentID INT PRIMARY KEY,
    Name VARCHAR(50)
);
```

Creating Child Table

```
CREATE TABLE Fees
(
    ReceiptID INT PRIMARY KEY,
    StudentID INT,

    FOREIGN KEY(StudentID)
    REFERENCES Student(StudentID)
);
```

Important Concepts

Can a Foreign Key Contain NULL?

Yes.
A Foreign Key may contain NULL, provided the column is defined to allow NULL values.
NULL usually means:

"This row is currently not related to any parent row."

Can a Table Have Multiple Foreign Keys?

Yes.
Example

Order

OrderID	CustomerID	EmployeeID
1	101	501

- CustomerID → Customer table

- EmployeeID → Employee table

One table can reference many other tables.

Can a Foreign Key Reference Itself?

Yes.
This is called a Self-Referencing Foreign Key or Recursive Foreign Key.
Example

Employee

EmployeeID	ManagerID
1	NULL
2	1
3	2

Here,
ManagerID is a Foreign Key that references EmployeeID in the same table.
This models hierarchical relationships such as:

- Employee → Manager
- Folder → Parent Folder
- Category → Parent Category

a foreign key is not always unique , many child rows may point to to same parent row.

Difference between Primary Key and Foreign Key?

Answer:
A Primary Key uniquely identifies rows within its own table and enforces Entity Integrity, whereas a Foreign Key references a key in another (or the same) table to establish relationships and enforce Referential Integrity.